

Establishment of antitick efficacy of a phytoformulation prepared from *Annona squamosa* leaf extracts for the management of acaricide resistant tick infestations on cattle

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ABSTRACT

To combat the problem of acaricide resistance in ticks, efforts have been made to develop eco-friendly herbal acaricides which are safe for animal use with no possibilities of evoking environment and residual toxicity. In the present study, the acaricidal properties of a commonly available plant, *Annona squamosa* have been established to develop a novel antitick phytoformulation using leaf extract, suitable solvents and emulsifier for managing resistant tick infestations in animals. The quality control of the safe and stable phytoformulation using Rutin as a major constituent was maintained by HPTLC profiling.

Several phytoformulations (A-E) were prepared and formulation 'C' is identified as the most effective showing 52.5–75% antitick activity against in vitro treated resistant strains of *R. microplus* (IVRI-IV and V) with 33.8–40.2% inhibition of oviposition while 65–85% mortality against multiacaricide resistant field populations. The phytoformulation provided 70.4% efficacy after first larval challenge in experimentally infested animals. The field efficacy of the formulation in different multilocal field trials was 68.53–77.8%. The detail antitick efficacy of the formulation along with safety and stability are explained as a component of antitick technology to manage resistant tick problem in livestock.

1. Introduction

Plants have been used as a rich source of medicines in veterinary and medical practices since long. In Ayurveda, approximately 2000 plant species are used for their medicinal importance. In the present era, besides allopathic medicines many plant based herbal medicines are used at a large scale throughout the world (Okeke et al., 2006; Tilburt and Kaptchuk, 2008; Karunamoorthi et al., 2012, 2013). About 750 medicinal plant species are used by traditional healers in the treatment of various diseases of animals and humans (Anjaria et al., 2000). The medicinal property of a plant can be improved by appropriate combination with other plant species or their active principles. The insecticidal, repellent, antifeedant, growth inhibitory, wound healing

properties, etc. of a number of plant species possessing a wide range of active phytochemicals have been reported (Isman, 2006; Cheraghi Niroumand et al., 2016; Bakele, 2018; Ahmed et al., 2020).

Annona squamosa (family: Annonaceae) is popularly known as sugar apple, custard apple or sweetsop and distributed throughout the tropical and subtropical parts of the world including India (Padmanabhan and Paliyath, 2016). Several alkaloids, flavonoids, quinones, tannins, anthocyanine, terpenoids, triterpenoids were abundantly present in bark, leaf and seeds of *A. squamosa* having antimicrobial, anti-inflammatory, hepatoprotective, anti-cancerous, anti-oxidant and insecticidal activity (Varadharajan et al., 2012; Gupta et al., 2014; Marahatta et al., 2019; Soni et al., 2020). Solvent guided extracts of various parts of *A. squamosa* were also reported to have mosquitocidal

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effect against *Culex* and *Anopheles* species (Kamaraj et al., 2011; Shad and Andrew, 2017; Akhila et al., 2021). Ilham et al. (2014) reported acaricidal activity of *A. squamosa* crude extract against *Hyalomma anatolicum*, a multi host tick species. Phytochemical and pharmacological studies on *A. squamosa* seeds have shown that the major bioactive compounds are annonaceous acetogenins, which have a strong anti-tumor activity (Chen et al., 2012). The hot water extract of *A. squamosa* leaves has been reported to possess hypoglycemic and antidiabetic activity (Gupta et al., 2005). Insecticidal activity of ethanolic extract was identified (Kumar et al., 2010) and the organic and aqueous extracts of *A. squamosa* leaf were found to have antibacterial activity (El-Chaghaby et al., 2014).

Ticks are voracious blood suckers, causing significant blood loss from animals resulting low productivity in terms of weight gain and milk yield, increased mortality and low quality hides. Among the species of ticks infesting cattle, *Rhipicephalus microplus* is the most widely distributed species and serves as the major vector of economically important pathogen like *Babesia* sp. and *Anaplasma* sp. (Ghosh et al., 2007). The use of synthetic acaricides of different classes is the mainstay in the tick management program for maintaining health status of livestock throughout the world. However, livestock owners reported the poor efficacy of many commonly used synthetic acaricides available in the market for tick management. Due to reduced efficacy, the animals are treated with commonly used acaricides without maintaining proper dose and treatment schedule (Bianchi et al., 2003). Consequently, the multi-acaricide resistant tick populations are established in Uruguay (Castro-Janer et al., 2010), Brazil (Andreotti et al., 2011; Klafke et al., 2017), North and South America, South Africa and Australia (Lovis et al., 2012), Uganda (Vudriko et al., 2016) and India (Nandi et al., 2018; Sagar et al., 2019). Besides, use of chemicals is adding environmental pollution and residual toxicity in milk and meat products.

All of these issues reinforce the need for alternative environment-friendly and sustainable approaches to control tick infestations. Ethno veterinary knowledge and practices are in use in many African and Asian countries for the management of ticks and other ectoparasites infesting livestock (Habeeb, 2010; Nabukenya et al., 2014; Ghosh et al., 2015; Dantas et al., 2015; Nchu et al., 2020). However, scientific validation of the knowledge is missing in most of the cases. A number of plant extracts have been screened for antitick activity using in vitro model and their efficacy has not been evaluated under in vivo conditions (Zaman et al., 2012; Ghosh et al., 2015; Valente et al., 2017; Islam et al., 2018; Upadhaya et al., 2020). Moreover, the information regarding antitick formulation with suitable delivery matrices is very scanty and thus effective natural formulation is not available commercially in most of the countries.

Previously, we conducted a pilot experiment to evaluate the efficacy of ethanolic extracts of seed of *A. squamosa* and 76.1% therapeutic efficacy was reported (Magadum et al., 2009). Taking the lead from our previous experiment, a detail investigation was conducted to validate the efficacy of the developed antitick formulation against experimental challenge and natural infestations conditions.

2. Materials and methods

2.1. Collection, processing and extraction of the plant materials

A. squamosa, commonly known as sweet apple, sugar apple, sweetsop or custard apple, is a small and well branched tropical tree which grows best in dry areas due to its high drought tolerance property. It bears small and sweet edible fruits. Plant materials i.e. leaves were collected in April and May from Lucknow, Uttar Pradesh, India, authenticated and herbaria were preserved (voucher specimen number 221,398) in CSIR-National Botanical Research Institute, Lucknow. The leaves were cleaned, shade dried and, coarsely powdered. The powder was soaked in 4 L of 50% hydro-ethanol (4 times x 4 L) at room temperature (30–35 °C) for overnight (18 h). The extract was decanted, mixed, filtered and

concentrated under reduced pressure (Buchi Rotavapor R-200, Switzerland) at 20–50 mm/Hg below 45 °C and lyophilized (Freeze dry system, Freezone 4.5, Labconco, England) (Ghosh et al., 2013). The prepared extract was kept at room temperature for further study.

2.2. Physicochemical & phytochemical analysis

Physicochemical analysis: Physicochemical parameters from the powdered plant material was determined and reported as loss on drying (LOD), total ash, acid insoluble ash, water soluble ash and solvent soluble extractive values (hexane, alcohol and water) (Peach and Tracy, 1955; World health organization 1998).

Preliminary phytochemical analysis: For qualitative estimation of tannins, total phenolics, flavanoids, sugar and starch contents present in the extracts the prescribed methods of Peach and Tracy (1955) and Singh et al. (2012) were adopted..

2.3. High performance thin layer chromatography (HPTLC)

HPTLC was performed on (20 cm × 10 cm) TLC aluminum plates precoated with 200-μm layer thickness of silica gel 60F254 (sd. Fine chem. Ltd, Mumbai, India, Product No. 25,438-P00) according to method of Tiwari et al. (2012). The extract samples were applied as 6 mm band width using Camag 100 microlitre sample syringe (Hamilton, Switzerland) with a Linomat 5 applicator (Camag, Switzerland) under a flow of N₂ gas. A range of mobile phase with varying polarity was tried but best suited linear ascending development was carried out with Ethyl acetate: n-butanol: Formic acid: Water (50: 30: 10: 10) as mobile phase in a Camag glass twin trough chamber (20 × 10 cm). The chamber was previously saturated with mobile phase vapor for 25 mins at room temperature (25 ± 2 °C) and plates were developed at a distance of approximately 80 mm from the point of application. After development, plates were dried through air drier and scanning was performed using Camag TLC Scanner 3 in UV absorbance mode operated by winCATS Software [Version 3.2.1]. The slit dimensions were 4 mm × 0.45 mm and the scanning speed was 100 mm/s.

2.4. In vitro bioassay for antitick activity

2.4.1. Reference tick strains

Acaricide susceptible tick strains: The colony of reference acaricide-susceptible IVRI-I strain of *R. microplus* (National Accession No. NBAII-IVRI-BM-1–1998) was maintained in the Entomology Laboratory, Division of Parasitology, Indian Veterinary Research Institute, Izatnagar. These ticks have not been exposed to any chemical acaricides to maintain their susceptibility and were used as the standard to evaluate acaricidal activity of the extracts.

Acaricide-resistant tick strains: A pyrethroid-resistant IVRI-IV (National Accession No. NBAIR-IVRI-BM-4–2009) and a multi-acaricide resistant IVRI-V of *R. microplus* were also maintained in the Entomology laboratory. The ticks of IVRI-IV colony were 194 fold resistant to deltamethrin while IVRI-V strain was 22.9, 20 and 25.4 fold resistant to deltamethrin, cypermethrin and diazinon respectively (Ghosh et al., 2017; Fular et al., 2018). These resistant tick strains were used to test the efficacy of the formulation following adult immersion test (AIT).

Field tick isolates: Tick isolates of *R. microplus* were collected from Bihar and Punjab states of India and were used to test the efficacy of the developed formulation. These ticks were characterized previously in a separate study and were resistant to deltamethrin, cypermethrin and diazinon and their resistance factor (RF) was determined as 3.07–34.9, 2.0–5.14 and 5.22–50.1, respectively (Kumar et al., 2011; Sharma et al., 2012).

The in vitro evaluation of the extracts was conducted by 2 min adult immersion test (Ghosh et al., 2011). The freshly collected engorged female ticks of IVRI-I strain of *R. microplus* were washed thoroughly, dried on tissue paper and used for testing antitick activity of the extracts. The

dose response assay of the extract using 2,4,6,8 and 10% (w/v) concentrations was conducted to determine LC_{50} and LC_{95} value (Ghosh et al., 2011). Briefly, twenty engorged females were used in each concentration which was replicated four times having five ticks in each replication ($N = 4 \times 5$). Each replication of ticks were immersed for 2 min in each concentration and then placed in Petri dishes padded with filter paper and kept at 28 °C and 85±5% RH. The ticks of the control group were immersed in distilled water as distilled water was used as a solvent in different concentrations of the extract. Ticks of both the experimental groups were maintained separately in the same temperature and humidity. The entire experiment was regularly observed and tick mortality was recorded up to 12–14 d until the control ticks completed oviposition. The inhibition of oviposition was calculated after weighing the eggs using the formula (Bennett, 1974).

$$RI = EW/IFW$$

$$\%IO = [(RI_{\text{control}} - RI_{\text{treated}}) / RI_{\text{control}}] \times 100$$

Where, RI = Reproductive Index; EW = egg weight; IFW = initial female weight; %IO = percentage inhibition of oviposition

2.4.2. Preparation of the formulation

To develop potential antitick phytoformulation, the extract (10% w/v) was mixed in water, added 1–2% tween 20 and 0.5–1.0% solvent by shaking it slowly until dark brown solution was formed. Five formulations were prepared using different concentrations of solvents (n-butanol) and surfactants (Tween 20) as mentioned below:

Formulation A = 10% extract + water

Formulation B = 10% extract + 2% tween-20 + water

Formulation C = 10% extract + 2% tween-20 + 0.5% solvent + water

Formulation D = 10% extract + 1% tween-20 + 1% solvent + water

Formulation E = 10% extract + 1% solvent + water

Each phytoformulation was tested against IVRI-I and IVRI-V strains by AIT as mentioned above and the best formulation was tested against reference pyrethroid resistant IVRI-IV strain and field isolates.

2.4.3. Bovine calves

Cross-bred male calves ($n = 20$) of 3–4 month age were procured from the Division of Livestock Production and Management, Indian Veterinary Research Institute, Izatnagar and maintained in a tick proof shed of the Division of Parasitology as per the guidelines of the regulatory authority, “Committee for the purpose of control and supervision on experimentation on animals” (Permission no. F.25/8/2016-CPCSEA).

2.4.4. Laboratory animals

Adult Wistar rats ($n = 30$) were kept at a temperature of 25 ± 3 °C, relative humidity in the range of 50–60% and a lighting sequence of 12 h light and 12 h dark. Rats were caged individually and provided with standard diet and ad libitum drinking water. Randomly selected rats were kept in their designated cages for 5d for acclimatization prior to dosing. The rats were selected in such a way that the weight difference was within $\pm 20\%$ of the mean weight. The necessary permission was obtained for use of animals in experimentation (permission no. 26–1/2015–16/J.D(R)/part file).

2.5. Safety and stability

The safety profile of the identified formulation C was carried out on Wistar rats following OECD guidelines - 410, 404 (OECD, 1981, 2002). The animals ($n = 18$, 6 rats in each group) of group I, II and III were given topical application with x, 2x and 5x of formulation ($x = 10\%$), respectively. The animals ($n = 12$) of group IV and V were treated with distilled water and vehicle, respectively. The skin thickness was recorded before and 24 h of application. Any visible changes at the site of

application of formulation, like staining, pruritis, and erythematic lesion were recorded at different time intervals before and after treatment. Respective control was maintained to compare the data. After completion of the experiment, the rats were sacrificed and the body weight, glutamic-oxalacetic transaminase (SGOT) and glutamic-pyruvic transaminase (SGPT), blood urea nitrogen (BUN), creatinine and hemoglobin were measured.

For stability study, the identified antitick phytoformulation C was kept at 2–4°, 25° and 37 °C temperature and the efficacy was checked periodically at monthly intervals up to eight months following AIT as mentioned earlier.

2.6. In vivo study

The calves were divided into two groups of six animals in each group (group I and II). Following ear bag method (Ghosh and Azhahianambi, 2007), each animal of both the groups was experimentally infested with 8–10 d old *R. microplus* larvae generated from 50 mg egg masses. The ear bags were checked daily to ensure the significant level of infestation. After 10–12 d of first larval challenge, the selected formulation was topically applied on the ears of calves of group I by swabbing while the calves of group II were treated with vehicle. The tick feeding bags were checked daily and entomological data on number of engorged tick dropped, mortality, percentage reduction of weight of adult females (DR.%), percentage of inhibition of oviposition (DO%) and percentage efficacy of the extract (E%) were calculated (Ghosh et al., 2013):

- i. $DT\% = 100 (1 - PNTV/PNTC)$, where DT% is the percent direct effect of extract on ticks; PNTV the mean number of engorged adults of treated group; and PNTC the mean number of engorged adults of control group.
- ii. $DR\% = 100 (1 - PMTV/PMTC)$, where DR (%) is the percentage reduction of mean weight of adult female; PMTV the mean weight of adult female of treated group; and PMTC the mean weight of adult female of control group.
- iii. $DO\% = 100 (1 - PATV/PATC)$, where DO (%) is the percentage reduction of mean weight of eggs; PATV the mean weight of eggs of treated group; and PATC the mean weight of eggs of control group.
- iv. $E\% = 100 [1 - (CRT \times CRO)]$, where E (%) is the efficacy of treatment on the survived ticks; CRT the reduction in number of adult females = $PNTV/PNTC$; and CRO the reduction in egg laying capacity = $PATV/PATC$.

To assess the duration of protection, second larval challenge was made on animals after 15 d of application.

2.7. Multilocational field trials

The field trials were conducted as per the guidelines of World Association for the Advancement of Veterinary Parasitology (WAAVP) (Holdsworth et al., 2006) with minor modifications. The level of tick infestation was estimated by counting the number of ticks in 6"x 6" patch at neck, thigh or ears and the counting was repeated after 15d of treatment to assess efficacy. The infestation level was classified as low (<50 ticks/patch), moderate (50–100 ticks/patch), high (>100 ticks/patch). Two hundred animals maintained in private dairy farms of three Indian states, Uttarakhand, Haryana and Rajasthan were selected on the basis of level of infestations. The data on types and frequency of acaricides used on the animals for tick control was recorded. Before conducting the trial engorged ticks were collected as per the protocol (Sharma et al., 2012) and were characterized using the established discriminating concentrations against deltamethrin and cypermethrin (Sharma et al., 2012), diazinon (Kumar et al., 2011) and ivermectin (Nandi et al., 2018).

For application of the phytoformulation swab method was adopted.

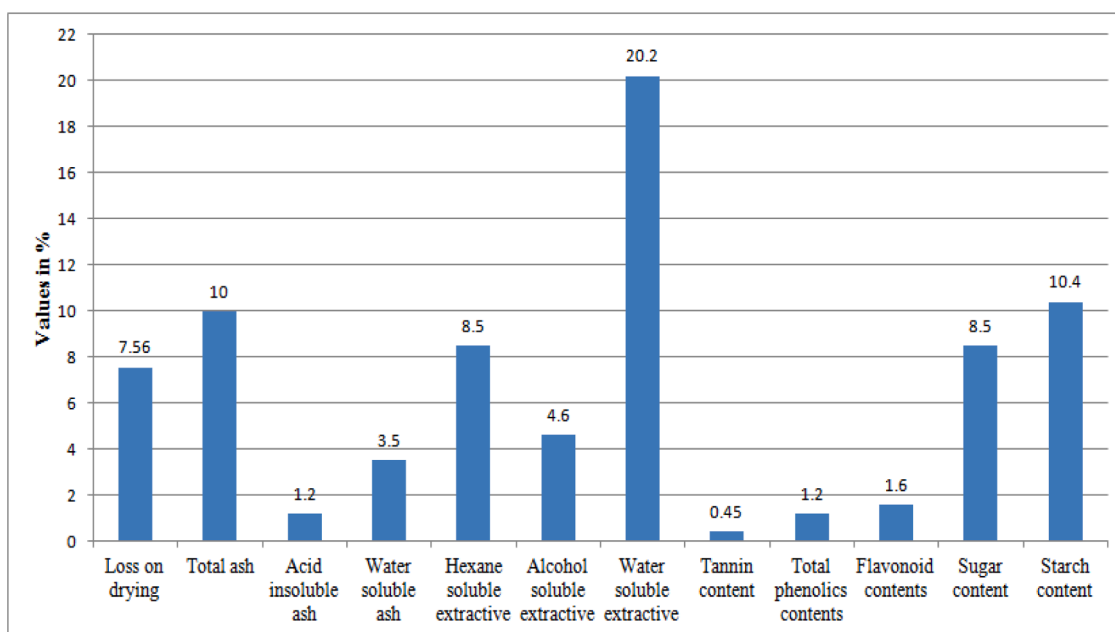


Fig. 1. Physicochemical and phytochemical analysis of hydroethanolic leaf extract of *A. squamosa*.

Post-treatment the animals were maintained as per the routine practice adopted by the respective farms. To assess the efficacy, the mean number of ticks per animal after 15 d of application was compared with the mean number of ticks per animal before application.

The percentage efficacy of the formulation was calculated by the following formula:

$$\text{Percentage field efficacy} = [1 - N_t / N_0] \times 100$$

Where, N_0 = Number of ticks per animal before application; N_t = Number of ticks after 15 d of treatment

2.8. Statistical analysis

Dose-response data of the extract and different formulations were subjected to probit analysis using GraphPad Prism 5.0 computer software (GraphPad Software, San Diego, California USA) to determine LC_{50}

and LC_{95} values (Finney, 1971). The significant difference in the mean values of treated and control group was determined by *t*-test. While the biochemical and hematological parameters were subjected to one-way analysis of variance (ANOVA) using Tukey test to determine significant difference in the mean values of treated, control and vehicle control groups. The significant difference in the mean values of entomological data of field isolates and vehicle control was estimated by one-way ANOVA using Dunnett test (Snedecor and Cochran, 1968).

3. Results

3.1. Physicochemical and phytochemical analysis of the extract

The results of total ash, acid-soluble and insoluble ash, water-soluble and insoluble ash, dry matter, and moisture content of leaves of *A. squamosa* are represented in Fig. 1. The total ash and acid insoluble

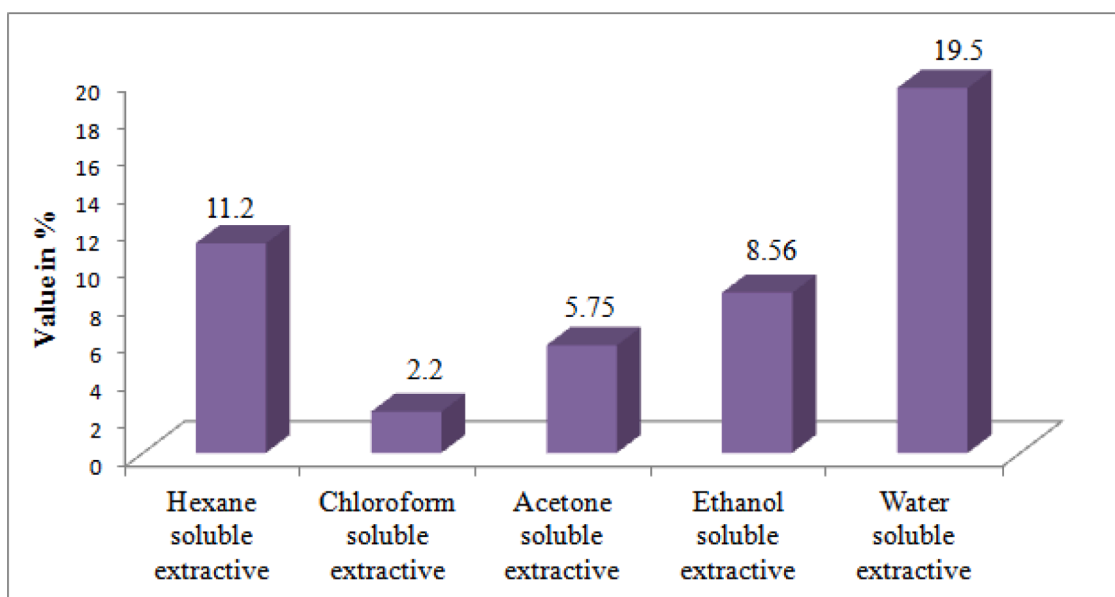


Fig. 2. Phytochemical analysis showing solvent soluble extractive values of hydroethanolic leaf extract of *A. squamosa*.

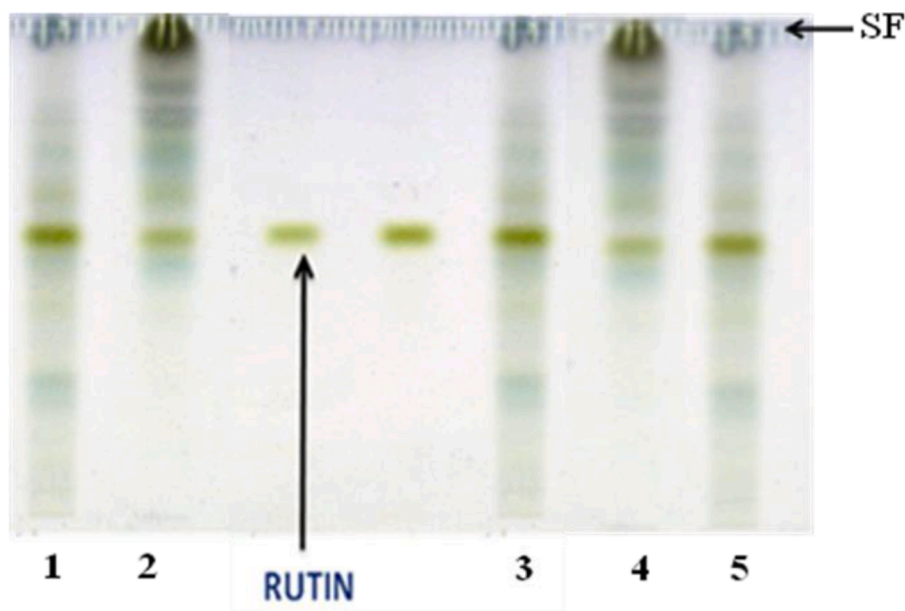


Fig. 3. HPTLC finger print profile of identified marker compound (Rutin) in different batches (1,2,3,4,5) of hydroethanolic (1:1) leaf extract of *A. squamosa*.

Table 1

Linear regression analysis of dose-response data of *Annona squamosa* leaf extract f on percent mortality and inhibition of oviposition (IO) against reference susceptible IVRI-I strain of *R. microplus*.

Variables (%)	Slope $\pm$ SE	R ²	LC ₅₀ (95% CI)	LC ₉₅ (95% CI)
Mortality	3.96 $\pm$ 0.52	0.893	4.41 (4.22 - 4.60)	10.6 (9.46–11.87)
IO	2.16 $\pm$ 0.14	0.972	3.29 (3.05 - 3.55)	13.6 (12.3–15.1)

ash are important parameter to identify the presence of inorganic substances. The present study revealed that the total ash and moisture content (Loss on drying) of leaves were 10 and 7.56%, respectively which were quite high. The water-soluble ash (3.50%) was higher than acid insoluble ash (1.20%). Extractive values are important parameter to check quality and variation in chemical constituents of the herbal drug. The water and alcohol soluble extractives, which are indicator of total solvent soluble component, were present at 20.2% and 4.6%, level, respectively (Fig. 1).

The phytochemical screening showed the presence of alkaloid, glycoside, tannin, triterpenoid, saponins and phenolic compounds in *A. squamosa* leaf extract. The water-soluble extractive value shows the presence of sugar, acids and inorganic compounds while alcohol soluble extractives indicates the presence of polar compounds like phenols, alkaloids, steroids, glycosides, flavonoids, etc. Quantitative estimation of primary metabolite showed the presence of tannins (0.45%), total phenolics (1.20%), flavonoid (1.60%), sugar (8.50%) and starch (10.40%). Successive extractive values indicated presence of hexane soluble (11.2%), chloroform soluble (2.2%), acetone soluble (5.75%), ethanol soluble (8.56%) and water soluble extractives (19.5%) (Fig. 2).

The HPTLC finger printing profiling of the extract revealed the presence of 'Rutin' as a major component (Fig. 3) and was used as a marker compound to maintain quality in different batches of the extracts prepared for the development of the phytoformulation.

3.2. In vitro efficacy

The hydroethanolic extract of *A. squamosa* leaf showed dose dependent mortality and inhibition of oviposition of reference susceptible IVRI-I strain of *R. microplus*. Probit analysis revealed LC₅₀ and LC₉₅ values (95% confidence intervals) of the extract were 4.41 (4.22 - 4.60) and 10.6% (9.46–11.87), respectively, for IVRI-I strain. The average

Table 2

Linear regression analysis of dose-response data of different phytoformulations from *A. squamosa* leaf extract against reference susceptible IVRI-I and multi-acaricide resistant IVRI-V strains of *R. microplus*.

Formulation	Tick Strain	Mortality slope ($\pm$ SE)	R ²	LC ₅₀ (95% Confidence Interval)	LC ₉₅ (95% Confidence Interval)
A	IVRI-I	15.20 $\pm$ 2.36	0.936	8.89 (8.77–9.01)	11.4 (11.1–11.7)
	IVRI-V	12.06 $\pm$ 1.50	0.956	8.97 (8.71–9.24)	12.27 (11.84–12.71)
B	IVRI-I	11.92 $\pm$ 2.26	0.903	8.24 (8.09–8.38)	11.3 (11.04–11.56)
	IVRI-V	8.90 $\pm$ 1.68	0.903	9.06 (8.09–10.15)	13.85 (11.17–17.17)
C	IVRI-I	8.75 $\pm$ 1.69	0.930	6.26 (6.11–6.41)	9.63 (8.52–10.88)
	IVRI-V	9.46 $\pm$ 2.07	0.875	7.45 (7.22–7.68)	11.10 (9.17–13.43)
D	IVRI-I	13.11 $\pm$ 1.84	0.962	7.48 (7.36–4.6)	9.97 (9.58–10.38)
	IVRI-V	12.24 $\pm$ 1.89	0.933	8.26 (8.83–10.0)	11.25 (10.42–12.15)
E	IVRI-I	12.19 $\pm$ 2.87	0.857	8.47 (8.3–8.64)	11.54 (11.27–11.82)
	IVRI-V	8.98 $\pm$ 2.91	0.761	8.86 (7.64–10.3)	13.48 (11.62–15.64)

weight of engorged females used in dose response assay was in the range of 113.5–118.7 mg and significant reduction in eggmass production (4.0–17.64 mg) of survived ticks as compared to control ticks (51.5 mg) was noted. At 10% concentration of the extract, the percentage inhibition of oviposition (%IO) was more than 80%. The LC₅₀ and LC₉₅ values of extract for percent IO were determined as 3.29 and 13.6% in the survived ticks (Table 1).

Five different phytoformulations were prepared using *A. squamosa* leaf extract, solvents, surfactant etc. and were tested. Comparing the LC₅₀ and LC₉₅ values of different formulations against reference susceptible IVRI-I and multiresistant IVRI-V strains, formulation C was selected as the most effective one with LC₉₅ value of 9.63 and 11.10%, and LC₅₀ of 6.26 and 7.45%, respectively (Table 2) which are minimum of all the five formulations. The formulation C was 52.5 $\pm$ 5.6% and 65–88% effective against reference resistant IVRI-IV strain and resistant

Table 3

Antitick activity of phytoformulation C from *A. squamosa* leaf extract against multiacaricide-resistant field isolates *R. microplus*.

Field isolates	% Mortality (Mean $\pm$ SE)	Reproductive Index (Mean $\pm$ SE)	% IO
MUZ	75.0 $\pm$ 9.6 ^c	0.310 $\pm$ 0.02 ^c	40.2
SUL	65.0 $\pm$ 5.8 ^c	0.319 $\pm$ 0.03 ^c	38.7
PAT	88.0 $\pm$ 6.0 ^c	0.238 $\pm$ 0.01 ^c	54.3
BTH	85.0 $\pm$ 9.6 ^c	0.312 $\pm$ 0.01 ^c	40.0
IVRI-IV	52.5 $\pm$ 5.6 ^b	0.344 $\pm$ 0.01 ^c	33.8
Control Vehicle	5.7 $\pm$ 3.7	0.520 $\pm$ 0.02	–

MUZ= Muzaffarpur, SUL= Sultanpur, PAT= Patiala, BTH=Bathinda, IVRI-IV= Pyrethroid resistant strain.

^b significant at $p < 0.01$,

^c significant at $p < 0.001$; significance was determined by one way ANOVA using Dunnett test.

Table 4

Biochemical parameters of rats treated with antitick phytoformulation from *A. squamosa* leaf extract.

Group	Dose (%)	BUN (mg%) (Mean $\pm$ SE)	Hb (gm %) (Mean $\pm$ SE)	SGPT (IU/L) (Mean $\pm$ SE)	SGOT (IU/L) (Mean $\pm$ SE)	Creatinine (mg/dL) (Mean $\pm$ SE)
I	10	13.5 $\pm$ 0.06	15.3 $\pm$ 0.16	47.3 $\pm$ 0.21	43.5 $\pm$ 0.14	0.66 $\pm$ 0.01
II	20	14.4 $\pm$ 0.11	16.0 $\pm$ 0.28	51.5 $\pm$ 0.17	45.5 $\pm$ 0.23	0.67 $\pm$ 0.0
III	50	13.4 $\pm$ 0.08	14.6 $\pm$ 0.16	55.4 $\pm$ 0.17	43.4 $\pm$ 0.12	0.75 $\pm$ 0.0
IV	Control	13.1 $\pm$ 0.03	15.6 $\pm$ 0.44	42.6 $\pm$ 0.08	47.4 $\pm$ 0.15	0.72 $\pm$ 0.0
V	Vehicle control	13.2 $\pm$ 0.02	14.8 $\pm$ 0.54	45.3 $\pm$ 0.12	46.2 $\pm$ 0.21	0.70 $\pm$ 0.01

Mean values of treated groups are insignificant at $p < 0.05$ level when compared with control and vehicle control.

field ticks, respectively (Table 3).

3.3. Safety and stability

The identified phytoformulation C was found safe. No significant change in biochemical parameters was observed in treated rats in comparison to control (Table 4). The formulation efficacy was stable up

to six months against treated ticks without any degradation in antitick property when stored at 2–4°, 25 °C and 37 °C temperature (Fig. 4).

3.4. In vivo efficacy

In vivo data indicated a significant reduction ($p < 0.001$) in the number of females successfully fed and dropped as engorged females (32.8 ± 7.4 to 37.7 ± 7.6) from the treated group of animals after first and second challenge in comparison to control group (85.0 ± 8.5 to 92.0 ± 8.5) of animals. The mean eggmass production of the engorged females dropped after first challenge was significantly ($p < 0.05$) limited to 56.5 ± 8.2 mg as against 72.5 ± 5.2 mg in ticks fed on control group of animals and accordingly, RI was also reduced. The DT% of ticks fed on treated animals after first challenge was 61.4% which was reduced to 59% after second challenge. Formulation also interfered with feeding of ticks and thus DR% was reduced up to 4.6% after second challenge (Table 5). Consequently, a 22.1% reduction in eggmass production (DO %) with an overall efficacy of 68–70.4% was noted.. The ticks attached with treated animals turned black with dark texture and sluggish after 3–4 d of application (Fig. 5).

3.5. Field trials

Field trials of identified phytoformulation C were conducted in organized and unorganized dairy farms of Rajasthan, Haryana and Uttarakhand in collaboration with veterinary officers, assistant professors and paravet officials.. Field efficacy was 68.53 ± 9.85 , 75.1 ± 6.5 and $77.8 \pm 4.3\%$ in trials conducted at Rajasthan, Haryana and Uttarakhand, respectively (Table 6, Fig. 6). The field efficacy of the formulation was highly satisfactory as reported by the animal owners, veterinary officers, assistant professors who were directly associated in the trials and supervised the animals.

4. Discussion

Due to overuse of chemical acaricides, the antitick property of commonly used acaricides has reduced significantly and finally multi-acaricide resistant tick populations is established in the field situation (Kumar et al., 2011, 2014; Sharma et al., 2012; Singh et al., 2014a, 2014b; Jyothimol et al., 2014; Nandi et al., 2018; Shakyia et al., 2020; Upadhyaya et al., 2020). To manage the problem of resistant ticks, a number of plants having antitick property have been identified. Phyto extracts are possessing multiple active principles present in different

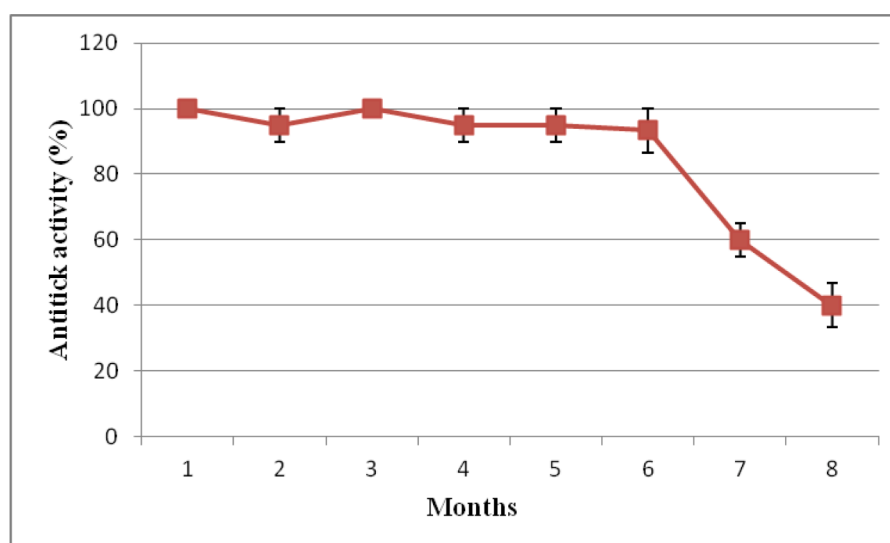
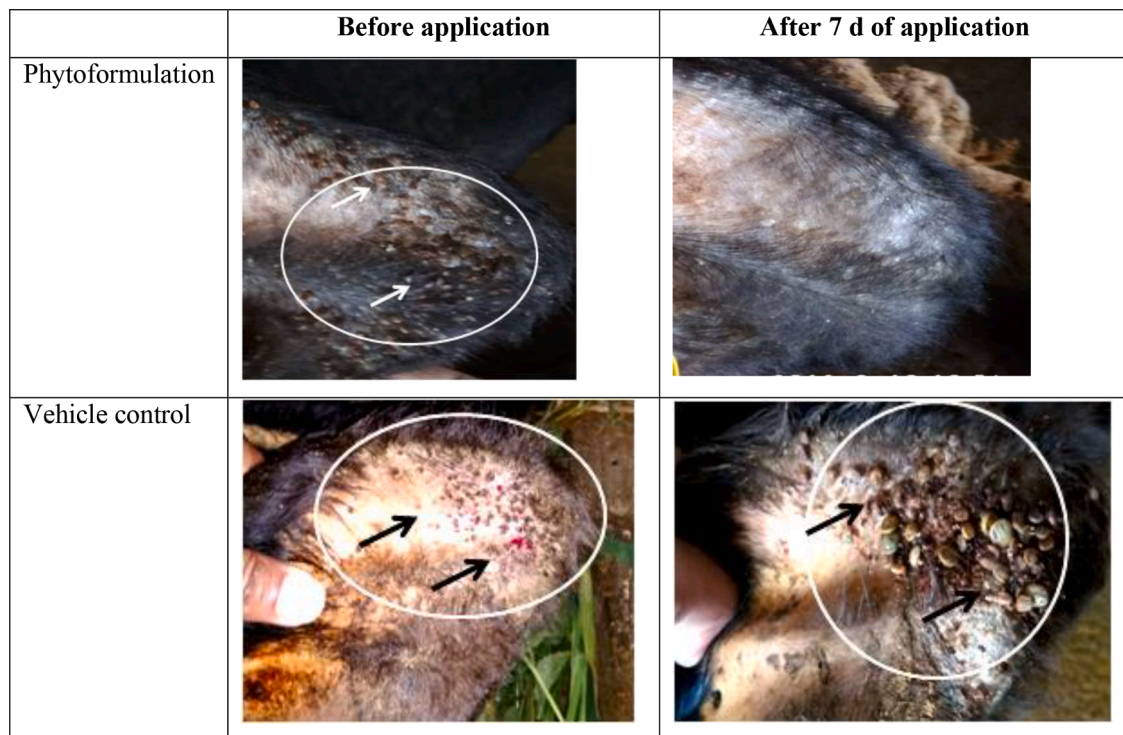


Fig. 4. Efficacy data of identified phytoformulation C against *R. microplus* stored at 60 °C.

Table 5In vivo efficacy of phytoformulation C applied on animals infested with *R. microplus*.

Efficacy on engorged adults dropped	Group	No. of engorged females dropped (Mean $\pm$ SD)	Tick wt. (mg) (Mean $\pm$ SD)	Mortality% (Mean $\pm$ SD)	Egg mass (mg) (Mean $\pm$ SD)	Reproductive Index (Mean $\pm$ SD)	Reduction in dropped engorged adults (DT%)	Reduction in engorged female wt. (DR.....%)	Reduction in egg wt. (DO %)	Overall efficacy (E %)
After first challenge	I	85.0 $\pm$ 8.5	137.1 $\pm$ 3.1	9.6 $\pm$ 2.5	72.5 $\pm$ 5.2	0.530 $\pm$ 0.03	–	–	–	–
	II	32.8 $\pm$ 7.4 ^c	122.5 $\pm$ 12.8	23.5 $\pm$ 15.4	56.5 $\pm$ 8.2 ^a	0.462 $\pm$ 0.04 ^a	61.4	10.6	22.1	70.4
After second challenge	I	92.0 $\pm$ 8.5	135.2 $\pm$ 10.3	10.7 $\pm$ 2.1	71.4 $\pm$ 9.7	0.526 $\pm$ 0.03	–	–	–	–
	II	37.7 $\pm$ 7.6 ^c	129.0 $\pm$ 10.6	15.5 $\pm$ 2.6	55.6 $\pm$ 9.1	0.430 $\pm$ 0.05	59.0	4.6	22.1	68.0

Group I = control, Group II = treated with extract, significant difference between the mean values of treated and control group was estimated using *t*-test..^a Significant at *p*<0.05,^c Significant at *p*<0.001.**Fig. 5.** In vivo efficacy of phytoformulation C in experimentally infested animals after 15 d of application.

ratios in a particular extract having different mode of action and are, working synergistically and thus chances of development of resistance to these is very low or rare (Chagas et al., 2003; Olivo et al., 2009; Habeeb, 2010; Zaman et al., 2012). Although a large number of acaricidal plant extracts have been tested against ticks under laboratory setup, only some of them have been evaluated on tick infested animals in order to validate the in vitro results (Zaman et al., 2012; Islam et al., 2018) and it was found that the efficacy was reduced under in vivo experiments (Ghosh et al., 2011, 2013).

The loss of in vivo antitick efficiency of plant extracts is undoubtedly a hindrance to develop alternative to acaricides and provided a scientific challenge before the researchers to develop an effective phytoformulation..

Chemoprofiling of plant extracts is one of the key components in the process of development of phytoformulation through which it is possible to identify marker compounds from active extracts which can be monitored throughout the development process and thus quality of the extracts/formulation can be maintained. The extract of *A. squamosa* was

found to have insecticidal activity due to the presence of annonin (Kumar et al., 2010), squamosin and acetogenins (Begum, 2013). Patel and Kumar (2008) identified five Linalool, Borneol, Eugenol, Farnesol, and Geraniol on the basis their Rf values in the aqueous and methanolic extracts of *A. squamosa* leaf and observed their antimicrobial activity. Similarly, the phytochemical analysis of hexane and methanol guided extracts of *Annona* leaf has unveiled alkaloids, polyphenols, flavonoids, and tannins and also reported its medicinal importance (Marahatta et al., 2019). In the present study the presence of alkaloid, glycoside, tannin, triterpenoid, saponins and phenolic compounds were detected. Previously, HPLC profiling of the ethanolic leaf extract of *A. squamosa* revealed the presence of Rutin, Quercetin, Kamepherol, Farmarixetin, and Isorhamnetin (Varadharajan et al., 2012) which also corroborates the present finding of rutin in the hydroethanolic (1:1) leaf extract.

Earlier, Chungsamarnyart et al. (1988) reported 90–100% larvicidal activity of *A. squamosa* ethanolic leaf extract against *R. microplus* while Ilham et al. (2014) reported both 100% larvicidal and adulticidal activity of crude leaf extract of *A. squamosa* in *Hyalomma anatolicum*.

Table 6

Field efficacy of phytoformulation C tested on tick infested animals stationed in organized and unorganized farms of Rajasthan, Haryana and Uttarakhand.

Particulars	Rajasthan	Haryana	Uttarakhand
No. of animals	94	46	70
Use of acaricides for tick control	Deltamethrin, cypermethrin	Deltamethrin, cypermethrin, ivermectin, organophosphates	Deltamethrin, ivermectin
Level of tick infestation* (animal number)	Low (28), Moderate (42), High (24)	Low (29), Moderate (11), High (6)	Low (24), Moderate (38), High (8)
Resistance status of ticks (RF)**	4.5 deltamethrin	4.5 deltamethrin, 4.2 diazinon	10.1 deltamethrin
Field efficacy (Mean% $\pm$ SE)	68.5 $\pm$ 9.8	75.1 $\pm$ 6.5	77.8 $\pm$ 4.3

*Low <50 ticks/patch, moderate 50–100 ticks/patch, high >100 ticks/patch (patch = 6"x6"); **determined in a separate study.

Similarly, the adulticidal activity of crude hydroethanolic leaf extract of *A. squamosa* against reference susceptible IVRI-I strain of *R. microplus* with an additional property of inhibition of egg production (Table 1) was obtained in the present study. The acaricidal activity of *A. squamosa* extracts was tested under in vivo and up to 71% protection against *R. microplus* challenge infestation was reported by Magadum et al. (2009). Kalakumar et al. (2000) investigated in vitro and in vivo activity of *A. squamosa* seed oil, neem oil and pyrethrins against *R. microplus*, *R. haemaphysaloides* and *H. anatolicum* and reported 100% efficacy of custard seed oil against buffalo ticks along with inhibition of oviposition.

Parate et al. (2014) tested formulations of aqueous extracts of five plants including *A. squamosa* in a particular combination and showed 100% mortality of *R. microplus* as compared to individual plant extract

under in vitro experiment while in the present study, the identified formulation C containing single plant extract showed more than 85% in vitro activity.

Data on safety and stability of natural formulations are the two vital components before conducting experiment on animals. In the present study, the identified formulation C was found stable at different thermal conditions of 2–4°, 25° and 37 °C for six months and was tested safe as per the OECD guidelines. However, storage of *M. azedarach* fruit extracts for five months at room temperature caused decrease in acaricidal effect (Sousa et al., 2008). The better stability of the formulation C observed in the present study may be due to the use of a specific solvent and detergent at a specific ratio with hydroethanolic extract. The aqueous leaf powder slurry of *A. squamosa* was reported to be nontoxic in acute toxicity study on Swiss albino mice as per OECD guidelines 420 (Sharma and Vd Goray Namrata, 2009). Similarly, Shivanna et al. (2019) reported no mortality, signs of toxicity and no significant elevations in hematological and biochemical blood parameters in animals treated with *Annona* species of plant and also advocated use of the plant species in pharmaceutical formulation.

Most of the plant based antitick extracts have been tested under in vitro model and the repeatability of in vitro generated data under in vivo conditions is not observed due to many factors like stability, solubility, delivery matrices, solvent systems etc. Ghosh et al. (2013) evaluated ethanolic extract of *R. communis* leaf extracts in experimentally infested animals and the in vivo efficacy was up to 59.9% after first challenge and, the data were comparatively less in comparison to the results obtained under in vitro (95 $\pm$ 5%) condition. Earlier, *Azadirachta indica* (Srivastava et al., 2008), *A. squamosa* (Magadum et al., 2009) and *Acorus calamus* (Ghosh et al., 2011) extracts were evaluated in experimentally infested animals and comparatively low efficacy of 47.1, 31.2 and 42.0%, respectively, was reported. However, in the present study, the efficacy of identified formulation C was 70.4% efficacious and was maintained upto 15 days of application. Zaman et al. (2012) developed a herbal formulation of 45% strength from aqueous extracts of *Azadirachta*

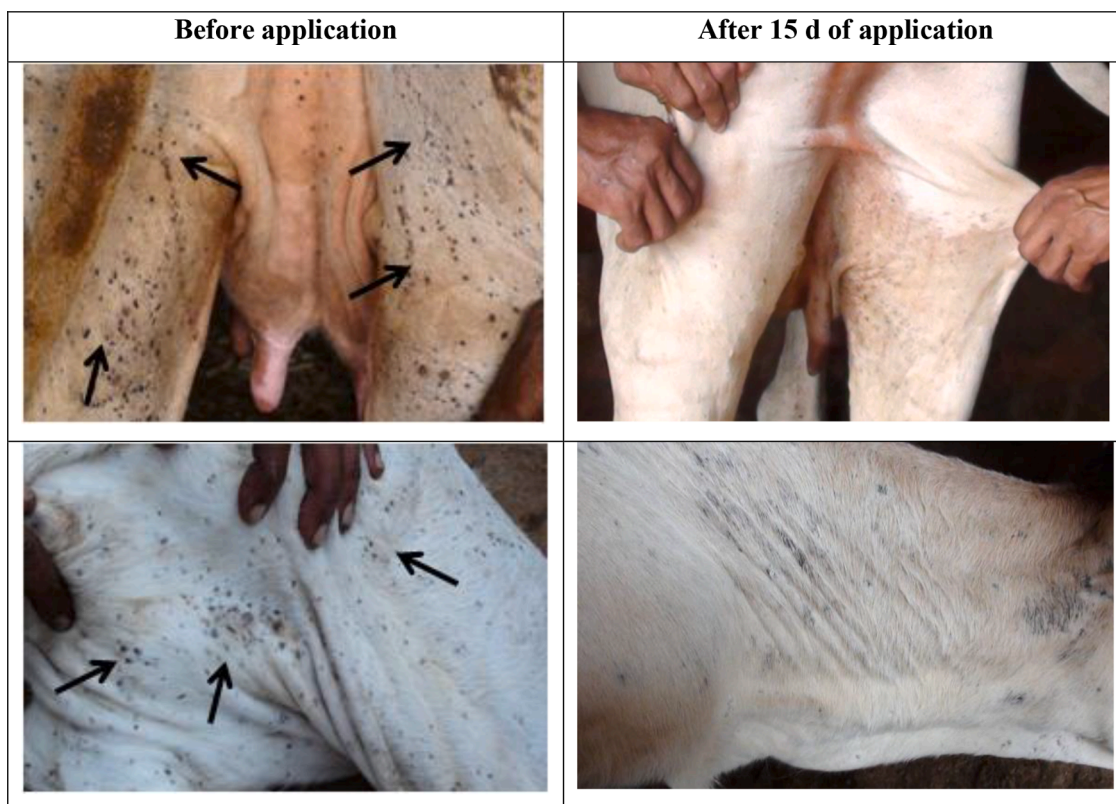


Fig. 6. Animals showing tick infestation before and after 15 days of application of phytoformulation C in multilocational trials.

indica leaves, *Nicotiana tabacum* leaves, *Calotropis procera* flowers and *Trachyspermum ammi* seeds and reported significant a reduction of 50.48% *R. microplus* ticks fed on the ears of treated animals as compared to negative control. However, in the present study, a significantly higher protection of 68.5 ± 9.8 to $77.8 \pm 4.3\%$ against natural infestation was noted even at a lower concentration of 10% formulation. As the efficacy data of the formulation was consistent in each stage of testing, we believe the formulation can provide a consistent efficacy against field tick infestation of animals irrespective of the resistant status of the tick species.

5. Conclusion

Analyzing the in vitro and in vivo antitick activity data of identified phytoformulation it is concluded that as the formulation possessing high acaricidal potential, may provide an alternative eco-friendly solution for the management of multiacaricide resistant tick infestations on animals..

Authors contribution

Anil Kumar Sharma, Shashi Shankar Tiwari, Sachin Kumar, Subha Rastogi conducted the experiment. Ajay Kumar Singh Rawat, Sharad Srivastava supervise the preparation of extracts and quality parameters. Debdatta Ray, Nirbhay Kumar Singh, Sumer Singh Rawat, A.K. Sangwan implement the multilocal field trials. Srikanta Ghosh arrange fund, design the experiment, monitored the experiment; Anil Kumar Sharma and Srikanta Ghosh drafted the ms.

Declaration of Competing Interest

The authors have stated that there is no conflict of interest associated with the publication and no financial support, which could have influenced the outcome.

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